Knowledge Organiser — 1.3 Exchanging Data

OCR A Level Computer Science (H446) — Component 01, Section 1.3 (Compression, encryption & hashing; Databases; Networks; Web technologies)

1.3.1 Compression, Encryption & Hashing

Compression = reduce file size → less storage, less bandwidth, faster transfer.

	Lossless	**Lossy**
Data removed permanently?	No	Yes
Original recoverable exactly?	Yes	No
Size reduction	Smaller	Larger
Uses	Text, code, ZIP, PNG	JPEG images, MP3 audio, video

- **Run Length Encoding (RLE):** lossless; replaces *runs* of repeated values with one value + a count. Best for long runs (simple images); can *enlarge* files with few repeats.
- **Dictionary compression:** lossless; stores recurring patterns once in a *dictionary* (index), replacing each with a short reference. Dictionary must be sent with the file. Good for text (e.g. LZW).

Encryption = scramble plaintext → ciphertext using algorithm + key; protects confidentiality if intercepted.

	Symmetric	**Asymmetric**
Keys	Same key encrypts & decrypts	Public + private key pair
Speed	Fast	Slower
Issue/benefit	Key distribution problem	Public key shared openly — solves distribution

- Asymmetric: data encrypted with **public** key → only matching **private** key decrypts. Enables digital signatures (sign with private, verify with public). HTTPS uses asymmetric to swap a symmetric session key, then symmetric for bulk data.

Hashing: hash function → fixed-length **hash/digest**. Properties: **one-way** (not reversible), **deterministic**, few **collisions**, fast. *Uses:* storing passwords (store hash, not password); hash tables (≈O(1) lookup). Hashing ≠ encryption (encryption is reversible with a key).

1.3.2 Databases

Flat file	**Relational**
Single table	Multiple linked tables
High redundancy	Reduced redundancy
Prone to anomalies/inconsistency	Consistent (normalised)
Simple	Scalable, complex queries

- **Entity** = thing data is stored about (→ table). **Attribute/field** = column. **Record/row/tuple** = one instance.

Keys | Key | Meaning | |---|---| | Primary | Attribute(s) uniquely identifying each record | | Foreign | Primary key of *another* table, used to link tables | | Composite | Primary key made of 2+ attributes combined | | Secondary | Non-primary attribute used to search/index quickly |

- **ER modelling:** relationships are 1:1, 1:M, M:N. A **many-to-many** is resolved with a **linking/junction table** (composite key).
- **Referential integrity:** a foreign key must always reference a valid existing primary key (or be null) → no orphaned records.

Normalisation = organise data to reduce redundancy & remove anomalies. | Form | Rule (in addition to previous) | |---|---| | 1NF | No repeating groups; atomic values; has a primary key | | 2NF | No **partial** dependencies (every non-key attribute depends on the *whole* key) | | 3NF | No **transitive** dependencies (no non-key attribute depends on another non-key) |

"The key, the whole key, and nothing but the key."

SQL keywords | Keyword | Use | |---|---| | `SELECT` | Choose which columns/fields to return | | `FROM` | Specify the table(s) | | `WHERE` | Filter which rows are returned (`AND`/`OR`, `LIKE 'A%'`) | | `JOIN ... ON` | Combine columns from related tables on a matching key | | `ORDER BY` | Sort results (`ASC`/`DESC`) | | `INSERT INTO ... VALUES` | Add a new record | | `UPDATE ... SET` | Change existing records | | `DELETE FROM` | Remove records | | `CREATE TABLE` | Define a new table |

Order: `SELECT ... FROM ... WHERE ... ORDER BY`.

Transactions & ACID — a transaction = one logical operation; all steps succeed or all fail. | Property | Meaning | |---|---| | **A**tomicity | All-or-nothing; failure → roll back, no partial change | | **C**onsistency | Moves DB from one valid state to another (keeps constraints) | | **I**solation | Concurrent transactions don't interfere | | **D**urability | Once committed, permanently saved (survives failure) |

- **Record locking:** lock a record during a transaction so others can't edit it → prevents **lost update problem**; risk of **deadlock**.
- **Redundancy (good):** duplicate data/hardware (backups, RAID, failover) for reliability — different from *data redundancy* (bad, removed by normalisation).

1.3.3 Networks

- **LAN:** small area, single site, one organisation owns hardware, high speed. **WAN:** large area (cities/countries), uses third-party infrastructure; internet = largest WAN.

Client-server	Peer-to-peer (P2P)
Server provides services to clients	All peers equal (client & server)
Central security/backup	Decentralised
Server = single point of failure	No single point of failure
Scales well, costlier	Cheaper, harder to manage/secure

Internet structure: *backbone* (high-capacity routes); *ISP* (provides access); *IP address* (uniquely identifies device; IPv4 32-bit, IPv6 128-bit); *DNS* (translates domain names → IP addresses, hierarchical & cached).

Topologies: Star — devices to central switch/hub; one cable fault isolated, but central node = single point of failure. **Mesh** — multiple paths between nodes; resilient/fault-tolerant, no single point of failure, but expensive (wired).

Packet switching	Circuit switching
Data split into packets, routed independently	Dedicated path for whole communication
Reassembled at destination	One fixed reserved path
Efficient, resilient (used by internet)	Guaranteed bandwidth, wastes idle capacity (phone)

- **Protocol** = rules for communication. **Standard** = agreed spec ensuring interoperability.

TCP/IP stack (data goes *down* when sending, *up* when receiving; each layer adds/removes a header) | Layer | Role | Protocols | |---|---|---| | Application | Services for user apps; prepares data | HTTP, HTTPS, FTP, SMTP, POP3, IMAP, DNS | | Transport | Splits into packets, end-to-end delivery, reassembly, ports | TCP, UDP | | Network/Internet | Adds source/destination **IP**, routes packets | IP | | Link/Data link | Physical connection, adds **MAC** address | Ethernet, Wi-Fi |

Protocols: HTTP = web pages; HTTPS = encrypted HTTP (TLS/SSL); FTP = file transfer; SMTP = *sending* mail; POP3 = retrieve mail (downloads & removes); IMAP = retrieve mail (stays on server, syncs).

Hardware: NIC (connects device, holds MAC); MAC address (unique 48-bit hardware address); **switch** (connects devices *within* a LAN, uses MAC); **router** (connects *different* networks, uses IP); WAP (wireless access to wired network).

Security: *firewall* (controls traffic by rules / packet filtering); *proxy* (intermediary — hides IP, caches, filters/logs); *encryption* (protects data in transit).

1.3.4 Web Technologies

Tech	Role
HTML	Structure & content (headings, paragraphs, links, images)
CSS	Presentation/style (colour, font, layout); selectors target tag, `.class`, `#id`
JavaScript	Interactivity/behaviour; usually client-side (events, validation, dynamic updates)

- **Search indexing:** web crawler/spider follows links, reads pages; engine builds an **index** mapping keywords → pages for fast retrieval, then ranks by relevance.
- **PageRank:** ranks pages by *importance* from incoming links. Each link = a "vote"; links from high-rank pages worth more; a page divides its rank across its outgoing links; **damping factor** (~0.85) models a random surfer; **iterative** until values **converge**.

	Client-side	Server-side
Runs on	User's browser	Web server
Speed	Fast, immediate	Slower (round trip)
Security	Insecure (can be viewed/bypassed)	Secure (hidden)
Server load	Reduced	Increased
Example	JS form validation	DB query, login/payment check

Must-know definitions

Term	Definition
Lossless / Lossy	No permanent data loss (recoverable) / permanent removal (not recoverable)
RLE	Lossless: replaces runs of repeated values with value + count
Dictionary compression	Lossless: stores repeated patterns once, references them
Symmetric / Asymmetric	Same key for both / public + private key pair
Hashing	One-way function producing a fixed-length hash
Primary / Foreign / Composite key	Uniquely IDs record / PK of another table linking tables / PK of 2+ attributes
Normalisation	Organising data to reduce redundancy & remove anomalies
Referential integrity	Foreign keys must reference a valid existing primary key
Transaction / ACID	All-or-nothing operation / Atomicity, Consistency, Isolation, Durability
Record locking	Locking a record during a transaction to stop concurrent edits
LAN / WAN	Small single-site network / large geographically spread network
Packet / Circuit switching	Packets routed independently / dedicated reserved path
Protocol	Set of rules governing communication
DNS	Translates domain names into IP addresses
MAC address	Unique hardware address of a NIC (link layer)
PageRank	Ranks pages by importance from incoming links

⭐ Top exam reminders

- **Lossy can NOT be reversed** — "reduced quality" ≠ "recoverable". RLE does **not** always shrink files (only with long runs).
- **Hashing ≠ encryption** — hashing is one-way; encryption is reversible with a key. Passwords are **hashed**, not encrypted.
- A **foreign key** is the **primary key of another table** — always name which table. For normalisation, name the dependency removed at each stage (repeating group → 1NF, partial → 2NF, transitive → 3NF) and explain *why* (remove redundancy/anomalies).
- **Switch** = devices within a LAN (MAC); **router** = different networks (IP). Don't swap them. Two meanings of "redundancy": data (bad) vs hardware/backup (good).

- **Client-side validation is NOT secure** — it can be bypassed; security-critical checks (login, payment) must also run **server-side**.